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still endemic in our country

Typhoid fever

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Definition

- An infectious feverish disease caused by the bacterium Salmonella typhi(*Salmonella enterica* Serovar Typhi) and less commonly by *Salmonella paratyphi*. also milder

- Acute generalized infection of the reticulo endothelial system, intestinal lymphoid tissue, and the gall bladder.

so human is the only reservoir

- The infection always comes from another human, either an ill person or a healthy carrier of the bacterium. The bacterium is passed on with water and foods and can withstand both drying and refrigeration.

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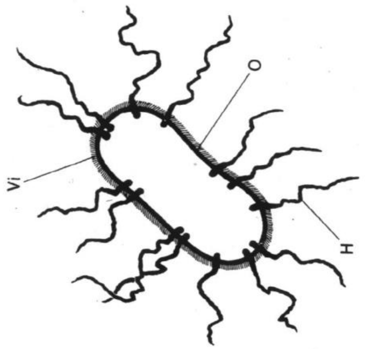
Causes

- Caused by the bacterium *Salmonella Typhi* .
- Ingestion of contaminated food or water. by human excretion
- Contact with an acute case of typhoid fever.
- Water is contaminated where inadequate sewerage systems and poor sanitation.
- Contact with a chronic asymptomatic carrier.
- Eating food or drinking beverages that handled by a person carrying the bacteria.
- *Salmonella enteriditis* and *Salmonella typhimurium* are other salmonella bacteria, cause food poisoning and diarrhoea.

Etiology :

- Typhoid fever is caused by a virulent bacterium called Salmonella typhi thriving in conditions of poor sanitation and crowding. G-ve bacilli in family Enterobacteriaceae



- Antigens: located in the cell capsule
  - ❖ H (flagellar antigen).
  - ❖ Vi (polysaccharide virulence Ag).
  - ❖ O (Somatic Ag)

but disease don't go from animal to human

Infects cattle, poultry, domestic cats, hamsters, humans etc.

- Refrigeration and freezing substantially slow or halt their growth. but when condition from growth become optimal it may grow again
- Pasteurizing and food irradiation kill *Salmonella* for commercially-produced foodstuffs containing raw eggs such as ice cream.

  no growing back
- Foods prepared in the home from raw eggs can spread salmonella if not properly cooked before consumption.

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How does the bacteria cause disease ?

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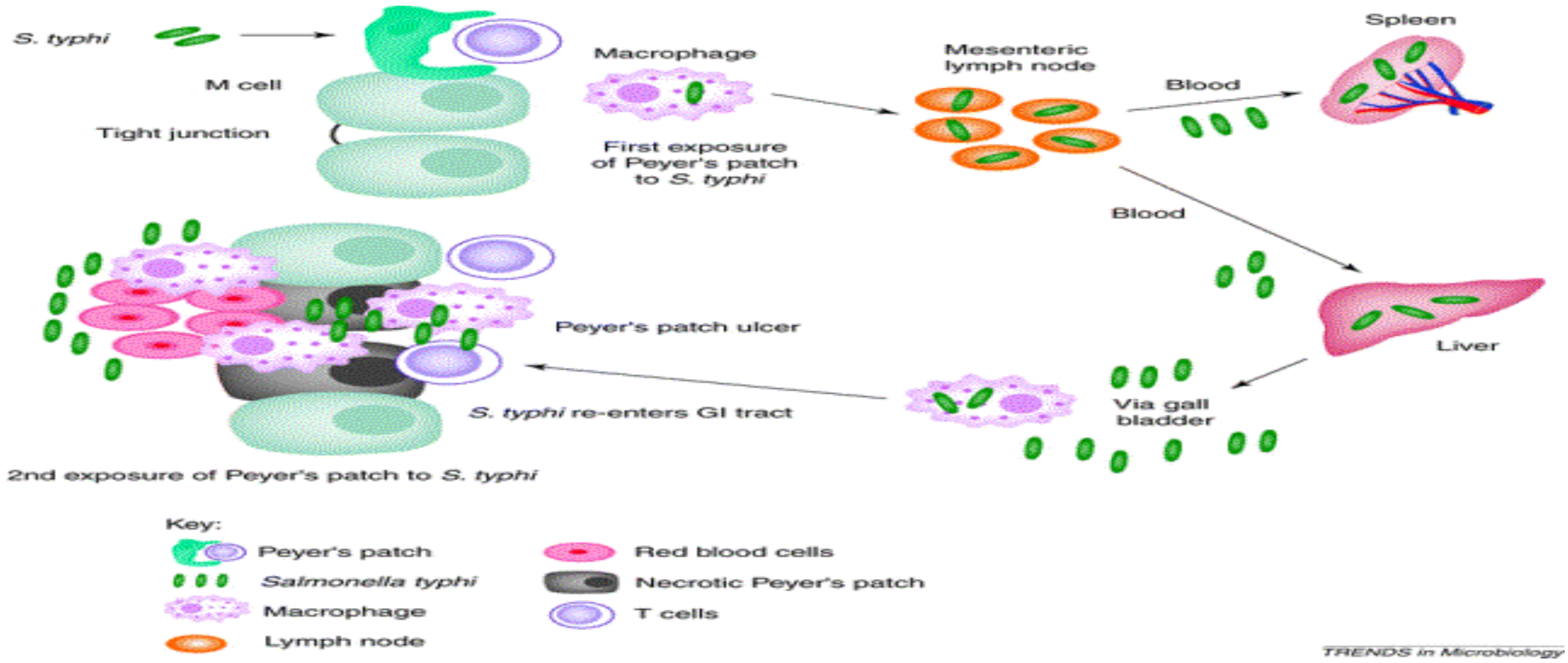

Ingestion of contaminated food or water

↓ *Salmonella* bacteria

Invade small intestine and enter the bloodstream

↓

Carried by white blood cells in the liver, spleen, and bone marrow

↓

Multiply and reenter the bloodstream

bacteria detected in blood culture within 1 week

↓

Bacteria invade the gallbladder, biliary system, and the lymphatic tissue of the bowel and multiply in high numbers

↓

Then pass into the intestinal tract and can be identified for diagnosis in cultures from the stool tested in the laboratory

↳ +ve after time usually 3 weeks

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Transmission

- *S typhi* has no non-human vectors.

→ so food handlers should recieve typhoid vaccinatio

1. via food handled by an individual who chronically sheds the bacteria through stool or, less commonly, urine
2. Hand-to-mouth transmission after using a contaminated toilet and neglecting hand hygiene
3. Oral transmission via sewage-contaminated water or shellfish

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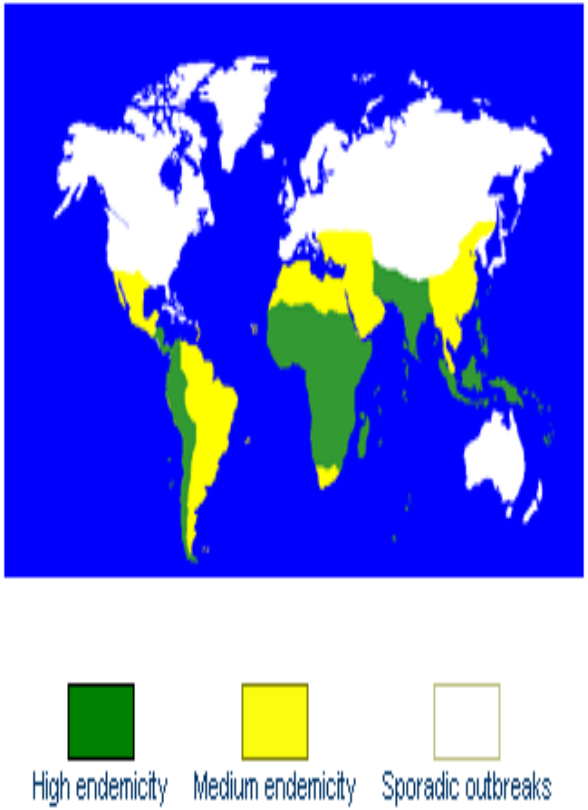
Epidemiology

- Typhoid fever occurs worldwide, primarily in developing nations whose sanitary conditions are poor.
- Typhoid fever is endemic in Asia, Africa, Latin America, the Caribbean, and Oceania.
- an estimated 9.2 million cases and 110,000 deaths occurring globally in 2019.

annually



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High Risk Groups

1. Worldwide, children are at greatest risk of getting the disease
2. Work in or travel to endemic area
3. Have close contact with someone who is infected or has recently been infected with typhoid fever
4. Weak immune system such as use of corticosteroids or diseases such as HIV/AIDS
5. Drinking water contaminated by sewage that contains S. typhi

Clinical presentation

- The incubation period for typhoid fever is 6-30 days.
- Illness onset is insidious, with gradually increasing fatigue and a fever that increases daily from low-grade to 39–40°C (approximately 102–104°F) by the third or fourth day of illness. Fever is commonly lowest in the morning, peaking in the late afternoon or evening. like TB
- If not treated, the symptoms develop over four weeks, with new symptoms appearing each week but with treatment, symptoms should quickly improve.

Clinical manifestations

The initial period (early stage due to bacteremia)

- **First week:** non-specific, insidious onset of fever

rising gradually

Fever up to 39-40^{0}C in 5-7 days, step-ladder(now seen in < 12%), headache chills, toxic, tired, sore throat, cough, abdominal pain and diarrhea or constipation.

vomiting

more common in children

usually no sweat

The fastigium stage

- second and third weeks.
- fever reaches a plateau at 39-40. Last 10-14 days.
- more toxic and anorexic with significant weight loss.
- The conjunctivae are injected, and the patient is tachypneic with a thready pulse and crackles over the lung bases.
- Abdominal distension is severe. Some patients experience foul, green-yellow, liquid diarrhea (pea soup diarrhea).

lack of interest (enthusiasm)

- **The(typhoid state) is characterized by apathy, confusion, and even psychosis. Necrotic Peyer patches may cause bowel perforation and peritonitis.**
- **At this point, overwhelming Toxaemia, Myocarditis, or Intestinal Haemorrhage may cause death.**

Signs and symptoms:

- **relative bradycardia.**
- **Splenomegaly, hepatomegaly**

less commonly seen due to aggressive management

- **rash (rose-spots) (occasionally) , maculopapular a faint pale color, slightly raised round or lenticular, fade on pressure, 2-4 mm in diameter, on the trunk, disappear in 2-3 days.**

Defervescence stage

- **By the fourth week of infection:**

If the individual survives , the fever, mental state, and abdominal distension slowly improve over a few days. Intestinal and neurologic complications may still occur. Weight loss and debilitating weakness last months. Some survivors become asymptomatic carriers and have the potential to transmit the bacteria indefinitely

convalescence stage

convalescence stage
• the fifth week: disappearance of all symptoms, but can relapse if not properly managed he still could be carrier

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Atypical manifestations :

➢Mild infection:

- ✓ very common seen recently
- ✓ symptom and signs are mild
- ✓ good general condition
- ✓ temperature is 38^0C
- ✓ short period of disease
- ✓ recovery expected in 1~3 weeks
- ✓ seen in early antibiotic users
- ✓ in young children more common
- ✓ easy to misdiagnose → causing under treatment → may relieve symptoms but increase risk of → relapse / → carrier

➢Persistent infection:

disease continue > 5 weeks

➢Ambulatory infection:

mild symptoms, early intestinal bleeding or perforation.

but has complication

- **Fulminant infection:**

rapid onset, severe toxemia and septicemia.

High fever, chill, circulatory failure, shock, delirium, coma, myocarditis, bleeding and other complications, DIC. => disseminated intravascular coagulation

- In the aged

temperature not high, weakness common.

More complications.

High mortality.

Complications

Intestinal bleeding or perforation

The most serious complication of typhoid fever

Other, less common

- Myocarditis
- Pneumonia
- pancreatitis
- UTI
- Osteomyelitis
- Meningitis
- Psychiatric problems

Complications

➢ Intestinal hemorrhage

✓ Commonly appear during the second-third week

✓ may be mild or severe bleeding

✓ often caused by unsuitable food, and diarrhea

✓ serious bleeding in about 2~8%

clues: sudden drop in temperature, rise in pulse, and signs of shock followed by dark or fresh blood in the stool. hypovolemic shock

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Blood cultures in Typhoid fever

- In Adults 5-10 ml of Blood is inoculated into 50 – 100 ml of Bile broth (0.5 %).
- Larger volumes 10-30 ml and clot cultures increase sensitivity
- Blood culture is positive as follows:

1st week in 90%

2nd week in 75%

3rd week in 60%

4th week and later in 25%

we use stool culture usually

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➢Bone marrow culture

the most sensitive test even in patients pretreated (up to 5 days) with antibiotics.

but invasive hence not 1st line

in here we maybe force since blood might be negative due to antibiotics

➢Urine and stool cultures

increase the diagnostic yield positive less frequently

stool culture better in 3rd~4th weeks

➢ Duodenal string test

to culture bile useful for the diagnosis of carriers. best for carrier but carriers can also be detected by stool but not by blood

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Widal test

- Serum agglutinins raise abruptly during the 2nd or 3rd week, it is +ve by 10th day, but max. during 18-23rd day
- The widal test detects antibodies against O and H antigens
  not bacteria
- Two serum specimens obtained at intervals of 7 – 10 days to read the rise of antibodies.
- The test is neither sensitive nor specific too much false +ve/-ve

Can't differentiate past & recent infection
(difficult to differentiate)

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TREATMENT

1-General :

- Isolation and rest
- suitable diet include easy digested food or half-liquid food and drinking more water
- IV fluid to maintain water and acid-base and electrolyte balance
- Symptomatic : antipyretic

Table 1 Treatment of Uncomplicated Typhoid.

duration no so important

Susceptibility	First-Line Oral Drug: Antibiotic	Daily Dose (mg/kg)	Days	Second-Line Oral Drug: Antibiotic	Daily Dose (mg/kg)	Days
Fully susceptible	Fluoroquinolone (e.g., ofloxacin)*	15	5–7†	Chloramphenicol	50–75	14–21
				Amoxicillin	75–100	14
				Trimethoprim–sulfamethoxazole	8 (trimethoprim)–40 (sulfamethoxazole)	14
Multidrug-resistant	Fluoroquinolone	15	5–7	Azithromycin	8–10	7
				Third-generation cephalosporin, e.g., cefixime	20	7–14
Quinolone-resistant‡	Azithromycin or fluoroquinolone	8–10 20	7 10–14	Third-generation cephalosporin, e.g., cefixime	20	7–14

Bactrim

Table 2 Treatment of Severe Typhoid.

Susceptibility	First-Line Parenteral Drug			Second-Line Parenteral Drug		
	Antibiotic	Daily Dose (mg/kg)	Days	Antibiotic	Daily Dose (mg/kg)	Days
Fully susceptible	Fluoroquinolone (e.g., ofloxacin)*	15	10–14	Chloramphenicol	100	14–21
				Ampicillin	100	10–14
				Trimethoprim–sulfamethoxazole	8 (trimethoprim)–40 (sulfamethoxazole)	10–14
Multidrug-resistant	Fluoroquinolone	15	10–14	Ceftriaxone or cefotaxime	60 80	10–14
Quinolone-resistant	Ceftriaxone or cefotaxime	60 80	10–14	Fluoroquinolone	20	10–14

Drug treatment

- ***Ciprofloxacin***: 15 mg/kg/d for 7 days
- For quinolone-resistant: ***azithromycin*** 10mg/kg/d for 7 days OR

  ceftriaxone 75mg/kg/d for 10-14 days

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steroids

- dexamethasone: initial dose 3 mg/kg by slow i.v. infusion over 30 minutes and after six hours, 1 mg/kg is administered and subsequently repeated at six-hourly intervals on seven further occasions, mortality can be reduced by some 80-90% in high-risk patients who has (high fever with obtundation and meningeal irritation signs)

reduce level of responsiveness and alertness

Carrier

Asymptomatic and have positive stool or rectal swab cultures for *S. typhi* a year following recovery from acute illness.

Treatment: **co-trimoxazole** 2 tab twice/d for 6 wk, OR

ciprofloxacin 750 mg twice/d for 4 wk

Carrier

Carriers should be excluded from activities involving food preparation and serving. Food handlers should not resume their duties until they have had three negative stool cultures at least one month apart.

- Vi Ab is used as a screening technique to identify carriers among food handlers and in outbreak investigations. Vi Abs are very high in chronic *S. typhi* carriers

Relapse

- **Apparent recovery can be followed by relapse in 5 – 10 % of untreated patient**
- **culture +ve of S.typhi after 1-3 wks of defervescence Symptom and signs reappear**
- **the bacilli have not been completely removed**
- **Some cases relapse more than once**
- On few occasions relapses can be severe and may be fatal.

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Prognosis: of typhoid fever

- Case fatality 0.5-1%.
- but high in old ages, infant, and serious complications
- immunity long lasting but not permanent
- About 3% of patients become fecal carriers .

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Vaccines for Typhoid Prevention

- Three types :

not used immunocomprised
4 dose given every other day

1. Oral – A live vaccine (Ty21a)

booster every 5 years

Polysaccharides not recommended below 2 years
liscened everywhere commonly used

single dose

2. Polysaccharide Typhoid Vaccine (Typhim–vi)

booster every 2-3 years
inactivated could be in immunocomprised

3- (TCV): Typhoid Conjugated Vaccine ****

not liscened everywhere but since conjugate could be used in less than 2 years

Indications

- High Risk Groups are:

1- workers in water projects

2- those who work in preparation of foods and drink (ice factory, mineral water).

3- swim pool workers

4- prisoners

5-Lab. Staffs

today then for tomorrow give day after

- Doses : 1- Typhium Vi (One I.M. Injection)

  2- Ty 21a (4 Oral Doses given each other day)

- Protection: 50-80 %

- Booster (2 Years for Typhium Vi

  (5 Years for Ty21a)

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→ Typhim-Vi → not recommended but if we have too much risk for other preventions to protect then give since mother's safety always outweights infant's

- Vaccination & pregnancy → Ty21a absolute contraindication

- Vaccine interaction (Antibiotics and Ty21a) → 72 hours space between them
  but Typhium could be used with anti-biotic

- Side Effect:

  swelling, pain, redness → especially site of injection

  Fever 10%

Paratyphoid fever

- It is similar in its symptoms to typhoid fever, but tends to be milder, with a lower fatality rate.
- It is caused by Paratyphi A, B, and C
- Rash may be more abundant
- May present as gastroenteritis specially in children

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Prophylaxis

Wash your hands.

Avoid drinking untreated water.

Avoid raw fruits and vegetables

Choose hot foods.